

12.3.3/ $\psi(\rho, \varphi) = A e^{-\frac{\rho^2}{2a^2}} \cos^2 \varphi \equiv R(\rho) Q(\varphi)$, where

$R(\rho)$, $Q(\varphi)$ normalize respectively. Since we only care about the angular momentum part, we can integrate the $R(\rho)$ part out, leaving only $Q(\varphi) = B \cos^2 \varphi$.

Now $Q(\varphi) = B \frac{1 + \cos 2\varphi}{2} = \tilde{B} \left[1 + \frac{e^{i2\varphi} + e^{i(-2\varphi)}}{2} \right]$

$= \sqrt{2\pi} \tilde{B} \left[\Phi_0 + \frac{1}{2} \Phi_2 + \frac{1}{2} \Phi_{-2} \right]$

$\Rightarrow P(l_z = m\hbar) = \frac{\left| \int_0^{2\pi} d\varphi \Phi_m^* Q(\varphi) \right|^2}{\int_0^{2\pi} d\varphi |Q(\varphi)|^2}$, Use $\int \Phi_m^* \Phi_{m'} d\varphi = \delta_{mm'}$, $\Rightarrow$
 $= 1$, just to make calculation simpler

$\Rightarrow P(l_z = 0) = \frac{|\sqrt{2\pi} \tilde{B}|^2 \cdot 1^2}{|\sqrt{2\pi} \tilde{B}|^2 \left(1^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right)} = \frac{2}{3}$

$P(l_z = 2\hbar) = P(l_z = -2\hbar) = \frac{|\sqrt{2\pi} \tilde{B}|^2 \cdot \left(\frac{1}{2}\right)^2}{|\sqrt{2\pi} \tilde{B}|^2 \left(1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right)} = \frac{1}{6}$

* Or from the expression of $Q(\varphi)$ we know

$\frac{P(l_z=0)}{P(l_z=2\hbar)} = \frac{P(l_z=0)}{P(l_z=-2\hbar)} = \frac{1}{\left(\frac{1}{2}\right)^2} = 4$, and $P(l_z=0) + P(l_z=2\hbar) + P(l_z=-2\hbar) = 1$

$\Rightarrow$ The same result

$$12.3.6/ \quad H = \frac{p^2}{2\mu} = -\frac{\hbar^2}{2\mu} \nabla^2 = -\frac{\hbar^2}{2\mu} \left(\vec{e}_\varphi \frac{1}{a} \frac{\partial}{\partial \varphi} \right)^2 = -\frac{\hbar^2}{2\mu a^2} \frac{\partial^2}{\partial \varphi^2}.$$

Here we used $\vec{\nabla} = \vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\varphi \frac{1}{a} \frac{\partial}{\partial \varphi}$ in polar coordinates. But because of the constraint, φ is the only coordinate any wave functions would depend on, so $\vec{\nabla} = \vec{e}_\varphi \frac{1}{a} \frac{\partial}{\partial \varphi}$.

On the other hand $L_z^2 = -\hbar^2 \frac{\partial^2}{\partial \varphi^2}$. It is then clear

$$(5) \quad H = \frac{L_z^2}{2\mu a^2}$$

We can use the eigenfunction of L_z :

$$H \Phi_m(\varphi) = \frac{L_z^2}{2\mu a^2} \Phi_m(\varphi) = \frac{m^2 \hbar^2}{2\mu a^2} \Phi_m(\varphi).$$

$$\Rightarrow E_m = \frac{\hbar^2}{2\mu a^2} m^2, \quad \text{degeneracy} = 2$$

$\sim \pm m$ ✓

$$12.3.8/ (1) \quad \vec{A} = \frac{B}{2} (-y \vec{i} + x \vec{j})$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = \frac{B}{2} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ -y & x & 0 \end{vmatrix} = \frac{B}{2} \cdot [\vec{i}(-\partial_z x) - \vec{j}(\partial_z y) + \vec{k}(\partial_x x + \partial_y y)]$$

(15)

$$= \frac{B}{2} \cdot \vec{k} \cdot 2 = B \vec{k}$$

(2) For a classical particle,

$$\mu \frac{v^2}{r} = q v B \Rightarrow v = \frac{q B r}{\mu}$$

$$\omega_0 = \frac{v}{2\pi r} = \frac{q B}{2\pi \mu}$$

If we change the unit $B \rightarrow B \frac{2\pi}{c}$, $\omega_0 = \frac{q B}{\mu c}$

$$(3) \quad H = \frac{(p_x + \frac{qB}{2c} y)^2}{2\mu} + \frac{(p_y - \frac{qB}{2c} x)^2}{2\mu}$$

$$Q = (p_x + \frac{qB}{2c} y)/qB, \quad P = (p_y - \frac{qB}{2c} x)$$

Clearly $[Q, Q] = [P, P] = 0$ because there is no non-vanishing commutators $[X, p_x] = [Y, p_y] = i\hbar$.

$$[Q, P] = \left[\frac{c p_x + \frac{qB}{2} y}{qB}, p_y - \frac{qB}{2c} x \right]$$

$$= \frac{c}{qB} \cdot \frac{qB}{2c} [p_x, -x] + \frac{1}{2} [y, p_y]$$

$$= i\hbar \Rightarrow Q, P \text{ are canonical.}$$

$$H = \frac{\left(\frac{qB}{c}Q\right)^2}{2\mu} + \frac{P^2}{2\mu} = \frac{q^2 B^2}{2\mu c^2} Q^2 + \frac{P^2}{2\mu}$$

$$\text{Let } \omega_0 = \frac{qB}{\mu c}, \quad H = \frac{P^2}{2\mu} + \frac{1}{2} \mu \omega_0^2 Q^2$$

This is the Hamiltonian for 1-D harmonic oscillator
with energy levels $E_n = (n + \frac{1}{2}) \hbar \omega_0$

* These levels are called Landau levels. And it shows that
(quantization)
in QM, energy levels get quantized even in some cases
where particles are free to travel into infinity, eg. $z \rightarrow \infty$
in this case.

$$\begin{aligned} (4) \quad H &= \frac{1}{2\mu} \left(P_x^2 + \left(\frac{qB}{2c}\right)^2 Y^2 + \frac{qB}{c} Y P_x + P_y^2 + \left(\frac{qB}{2c}\right)^2 X^2 - \frac{qB}{c} X P_y \right) \\ &= \frac{P_x^2 + P_y^2}{2\mu} + \frac{\mu \omega_0^2}{8} (X^2 + Y^2) - \frac{\omega_0}{2} L_z \\ &= H\left(\frac{\omega_0}{2}, \mu\right) - \frac{\omega_0}{2} L_z \end{aligned}$$

$= (X P_y - Y P_x)$
 $\omega_0 = \frac{qB}{\mu c}$

where $H\left(\frac{\omega_0}{2}, \mu\right)$ is the Hamiltonian for isotropic
2-D oscillator with frequency $\frac{\omega_0}{2}$

$$\begin{aligned}
[H(\frac{\omega_c}{2}, \mu), L_z] &= [H(\frac{\omega_c}{2}, \mu), X P_y - Y P_x] \\
&= \frac{1}{2\mu} ([P_x^2, X P_y] + [P_y^2, -Y P_x]) \\
&\quad + \frac{\mu v_0^2}{8} ([X^2, -Y P_x] + [Y^2, X P_y]) \\
&= \frac{1}{2\mu} (2 P_x P_y (-i\hbar) + 2 (i\hbar) P_y P_x) \\
&\quad + \frac{\mu v_0^2}{8} (2 Y X (-i\hbar) + 2 X Y (i\hbar)) \\
&= 0 \quad ([P_x, P_y] = [X, Y] = 0)
\end{aligned}$$

$\Rightarrow$ For $H(\frac{\omega_c}{2}, \mu)|n\rangle = \frac{1}{2}\hbar(\frac{\omega_c}{2})(n+1)|n\rangle$, where

$|n\rangle$ is the basis diagonalizing $H(\frac{\omega_c}{2}, \mu)$,

$$H(\frac{\omega_c}{2}, \mu)(L_z|n\rangle) = L_z H(\frac{\omega_c}{2}, \mu)|n\rangle = \hbar(\frac{\omega_c}{2})(n+1)(L_z|n\rangle)$$

$\Rightarrow L_z|n\rangle$ is also an eigenfunction of $H(\frac{\omega_c}{2}, \mu)$
with quantum number n .

$\Rightarrow L_z|n\rangle = \tilde{k}|n\rangle$. This is just the
eigenfunction for L_z , so we know $\tilde{k} \equiv L_z = m\hbar$,
where m is another quantum number specifying
the degeneracy in $|n\rangle$ (Recall $n = n_x + n_y$).
See exercise 10.2.2

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$$\Rightarrow H|n\rangle = \left(H \left(\frac{\omega_c}{2}, \mu \right) - \frac{\omega_c}{2} L_z \right) |n\rangle$$

$$= \left(\hbar \frac{\omega_c}{2} (n+1) - \frac{\omega_c}{2} m \hbar \right) |n\rangle$$

So $|n\rangle$ also diagonalizes H . The energy levels are

$$E_{n,m} = \frac{\hbar \omega_c}{2} (n+1) - \frac{\hbar \omega_c}{2} m$$

Take $n \equiv 2k + |m|$, $E = \hbar \omega_c \left(k + \frac{|m|}{2} - \frac{m}{2} + \frac{1}{2} \right)$

For $m \geq 0$, $E = \hbar \omega_c \left(k + \frac{1}{2} \right)$

$m < 0$, $E = \hbar \omega_c \left(k - m + \frac{1}{2} \right)$

Thus, levels are the same as $E = \hbar \omega_c \left(n + \frac{1}{2} \right)$

12.4.3/(1) According to the picture,

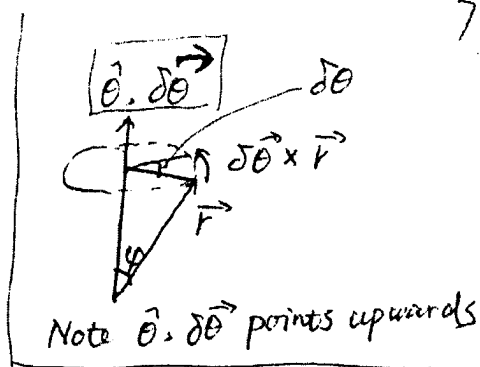
when $\vec{r}$ is rotated by an angle $\delta\theta$

along $\hat{\theta}$ (denote $\delta\vec{\theta} \equiv \delta\theta \hat{\theta}$), it

moves ~~to~~ the distance of $r \sin\phi \delta\theta$ along the

direction ~~to~~ $\hat{\theta} \times \vec{r}$. Thus, we can summarize it as

$$\vec{r} \rightarrow \vec{r} + \delta\vec{\theta} \times \vec{r}$$



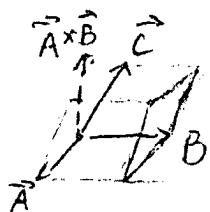
$$(2) \quad U[R(\delta\vec{\theta})] \psi(\vec{r}) = (I - \frac{i}{\hbar} \delta\theta L_{\hat{\theta}}) \psi(\vec{r})$$

$$= \psi(\vec{r}) - \frac{i}{\hbar} \delta\theta L_{\hat{\theta}} \psi(\vec{r}) \quad (1)$$

Besides, $U[R(\delta\vec{\theta})] \psi(\vec{r}) = \psi(\vec{r} - \delta\vec{\theta} \times \vec{r})$

$$= \psi(\vec{r}) - (\delta\vec{\theta} \times \vec{r}) \cdot \vec{\nabla} \psi(\vec{r}).$$

Use the identity: $(\vec{A} \times \vec{B}) \cdot \vec{C} = \vec{A} \cdot (\vec{B} \times \vec{C})$



(This represents the volume of a "leaning" cube with $\vec{A}$, $\vec{B}$, $\vec{C}$ as three edges)

$$\Rightarrow U[R(\delta\vec{\theta})] \psi(\vec{r}) = \psi(\vec{r}) - \delta\vec{\theta} \cdot (\vec{r} \times \vec{\nabla}) \psi(\vec{r})$$

$$= \psi(\vec{r}) - \frac{i}{\hbar} \delta\theta (\hat{\theta} \cdot (\vec{r} \times \vec{\nabla})) \psi(\vec{r})$$

$$= \psi(\vec{r}) - \frac{i}{\hbar} \delta\theta (\hat{\theta} \cdot \vec{L}) \psi(\vec{r}) \quad (2)$$

$$(1) (2) \Rightarrow L_{\hat{\theta}} = \hat{\theta} \cdot \vec{L}$$

$$12.5.3/\# J_{\pm} = J_x \pm iJ_y$$

$$\Rightarrow \begin{cases} J_x = \frac{J_+ + J_-}{2} & J_x^2 = \frac{J_+^2 + J_-^2 + J_+ J_- + J_- J_+}{4} \\ J_y = \frac{J_+ - J_-}{2i} & J_y^2 = \frac{\cancel{J_+^2 - J_-^2 - J_x^2 - J_y^2} - J_+^2 - J_-^2 + J_+ J_- + J_- J_+}{4} \end{cases}$$

(1) Since $J_{\pm}|j, m\rangle \rightarrow |j, m\pm 1\rangle$, we have

$$\langle J_{\pm} \rangle \equiv \langle jm | J_{\pm} | jm \rangle = 0, \text{ and therefore}$$

$$\langle J_x \rangle = \langle J_y \rangle = 0$$

$$(2) \quad \langle J_x^2 \rangle = \frac{1}{4} \langle jm | (J_+ J_- + J_- J_+) | jm \rangle$$

Here we have

$$\begin{aligned} J_+ J_- &= (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2 + i(J_y J_x - J_x J_y) \\ &= J^2 - J_z^2 + \hbar J_z \\ J_- J_+ &= (J_x - iJ_y)(J_x + iJ_y) = J^2 - J_z^2 - \hbar J_z \end{aligned}$$

$$\begin{aligned} \Rightarrow \langle J_x^2 \rangle &= \frac{1}{4} \langle jm | 2J^2 - 2J_z^2 | jm \rangle = \frac{1}{2} [j(j+1)\hbar^2 - m^2\hbar^2] \\ &= \frac{\hbar^2}{2} (j(j+1) - m^2) \end{aligned}$$

$$\langle J_y^2 \rangle = \frac{1}{4} \langle jm | (J_+ J_- + J_- J_+) | jm \rangle = \langle J_x^2 \rangle$$

$$(3) \quad \Delta J_x = \Delta J_y = \sqrt{\langle J_x^2 \rangle} = \hbar \sqrt{\frac{1}{2} [j(j+1) - m^2]}$$

$$\Delta J_x \cdot \Delta J_y = \frac{\hbar^2}{2} [j(j+1) - m^2]$$

$$\begin{aligned} \text{From (9.2.9), } (\Delta J_x)^2 (\Delta J_y)^2 &\geq |\langle j_m | J_x J_y | j_m \rangle|^2 \\ &= |\langle j_m | \frac{J_+ J_- - J_- J_+}{4i} | j_m \rangle|^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow \Delta J_x \cdot \Delta J_y &\geq |\langle j_m | \frac{2\hbar J_z}{4} | j_m \rangle| = \frac{\hbar}{2} |m| \hbar \\ &= \frac{|m|}{2} \hbar^2 \end{aligned}$$

$$\text{We can see that } [j(j+1) - |m|^2] - |m| = j(j+1) - |m|(|m| + 1)$$

$$\text{Since } |m| \leq j, \quad j(j+1) - |m|(|m| + 1) \geq 0$$

$$\Rightarrow \Delta J_x \cdot \Delta J_y = \frac{\hbar^2}{2} [j(j+1) - m^2] \geq \frac{|m|}{2} \hbar^2 \text{ is satisfied}$$

$$(4) \quad \text{When } m = \pm j \Rightarrow$$

$$\Delta J_x \Delta J_y = \frac{\hbar^2}{2} [j(j+1) - j^2] = \frac{\hbar^2}{2} j = \frac{\hbar^2}{2} |m|, \text{ the}$$

inequality is saturated.

$$12.5.13/4) \quad Y_{\pm 1}^{\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin\theta e^{\pm i\varphi}$$

$$= \mp \sqrt{\frac{3}{8\pi}} \frac{1}{r} [r \sin\theta \cos\varphi \pm i r \sin\theta \sin\varphi]$$

$$= \mp \sqrt{\frac{3}{8\pi}} \frac{x \pm iy}{r}$$

$$\begin{cases} Y_1^1 = \sqrt{\frac{3}{8\pi}} \frac{-x-iy}{r} \\ Y_1^{-1} = \sqrt{\frac{3}{8\pi}} \frac{x-iy}{r} \end{cases}$$

$$Y_0^0 = \sqrt{\frac{3}{4\pi}} \cos\theta = \sqrt{\frac{3}{4\pi}} \frac{z}{r}$$

$$(2) \quad \psi = N(x+y+2z) e^{-\alpha r}$$

Here $\begin{cases} x = (Y_1^{-1} - Y_1^1) \sqrt{\frac{2\pi}{3}} r \\ y = i(Y_1^{-1} + Y_1^1) \sqrt{\frac{2\pi}{3}} r \end{cases}$

$$z = Y_0^0 \sqrt{\frac{4\pi}{3}} r$$

$$\Rightarrow \psi = N e^{-\alpha r} \left[\sqrt{\frac{2\pi}{3}} r (Y_1^{-1}(1+i) - Y_1^1(1-i)) + 2\sqrt{\frac{4\pi}{3}} r Y_0^0 \right]$$

$$\Rightarrow P(l_3=0) = \frac{\int d\Omega |Y_0^0 \psi|^2}{\int |4|^2 d\Omega} = \frac{(2\sqrt{2})^2}{(2\sqrt{2})^2 + |1+i|^2 + |1-i|^2} = \frac{2}{3}$$

* Here we drop the integration over r because this variable can be separated in ψ and this part is integrated into unity in both numerator & denominator.

$$\text{Similarly } P(l_3=\pm\hbar) = \frac{\int d\Omega |Y_{\pm 1}^{\pm 1} \psi|^2}{\int |4|^2 d\Omega} = \frac{|1\pm i|^2}{(2\sqrt{2})^2 + |1+i|^2 + |1-i|^2} = \frac{1}{6}$$

$$\rightarrow * \psi = N e^{-\alpha r} \sqrt{\frac{2\pi}{3}} r [Y_1^{-1}(1+i) - Y_1^1(1-i) + 2\sqrt{2} Y_0^0]$$